PiP-LU DAY

November 5, 2026

Ever wondered what your colleague down the hall is actually working on?

PiP-LU Day brings together postdoctoral researchers and junior PIs to connect, share their research, and explore new collaborations. Through short talks, speed-networking sessions and informal discussions, the day strengthens interactions across the department, raises the visibility of ongoing projects, and helps Master's students find research projects and future career paths.

DATE	VENUE	WHO
Thursday 5 November 2026	Ekologihusets stora entréhall	Junior PIs and postdocs

COMMENT

PiP-LU Day is an initiative of the PiP committee, supported by the Research Council of the Department of Biology and the Research Council of the Department of Earth and Environmental Sciences (MGEO) at Lund University. Anyone interested can attend the presentations, including the Topic Discussions, and meet the sponsors.

Lund University

Short programme

SciLifeLab have a stand in the entrance hall all day, and SBDI / GBIF Sweden until 15:00 — all five SciLifeLab units join over lunch.

08:15 – 08:45	**Registration**	Badges and arrival in the entrance hall
08:45 – 09:00	**Introduction to the meeting**	Welcome from the organising committee
09:00 – 10:30	**Research session I**	6 talks · 10 min each + 2 min questions
10:30 – 11:15	**Networking session**	Two 20 min rounds · rotating groups of six · rooms on your badge
11:15 – 12:30	**Research session II**	5 talks · 10 min each + 2 min questions
12:30 – 13:30	**Lunch** SBDI · GBIF SWEDEN SCILIFELAB	All five SciLifeLab units at the stand
13:30 – 14:30	**Topic discussion**	Funding opportunities first, then student supervision and teaching
14:30 – 15:00	**Coffee break** SBDI · GBIF SWEDEN SCILIFELAB	
15:00 – 16:15	**Research session III**	5 talks · 10 min each + 2 min questions
16:15 – 17:15	**Master meeting – Find your project**	Project offers at a relaxed poster-fair mingle
17:15 – 18:15	**Research session IV**	4 talks · 10 min each + 2 min questions
18:15 – 18:30	**Closing remarks**	Raffle draw and thanks to our supporters
18:30 →	**Finger food buffet and drinks**	Open networking to close the day

What to expect?

The four session formats, and what to expect from each.

RESEARCH SESSIONS

Short research talks where postdocs and junior PIs present their research interests. Speakers are also encouraged to briefly introduce their research journey and career path, offering insight into the different routes that can lead to independent research. Each talk is 10 minutes plus 2 minutes of questions, with a short changeover before the next speaker.

NETWORKING SESSION

A relaxed speed-networking session run as two 20-minute rounds. You sit in a small group of about five people, and the groups rotate between rounds so you meet a fresh set of colleagues — a chance to get to know each other, share research interests, exchange ideas, and maybe spark new collaborations.

TOPIC DISCUSSION

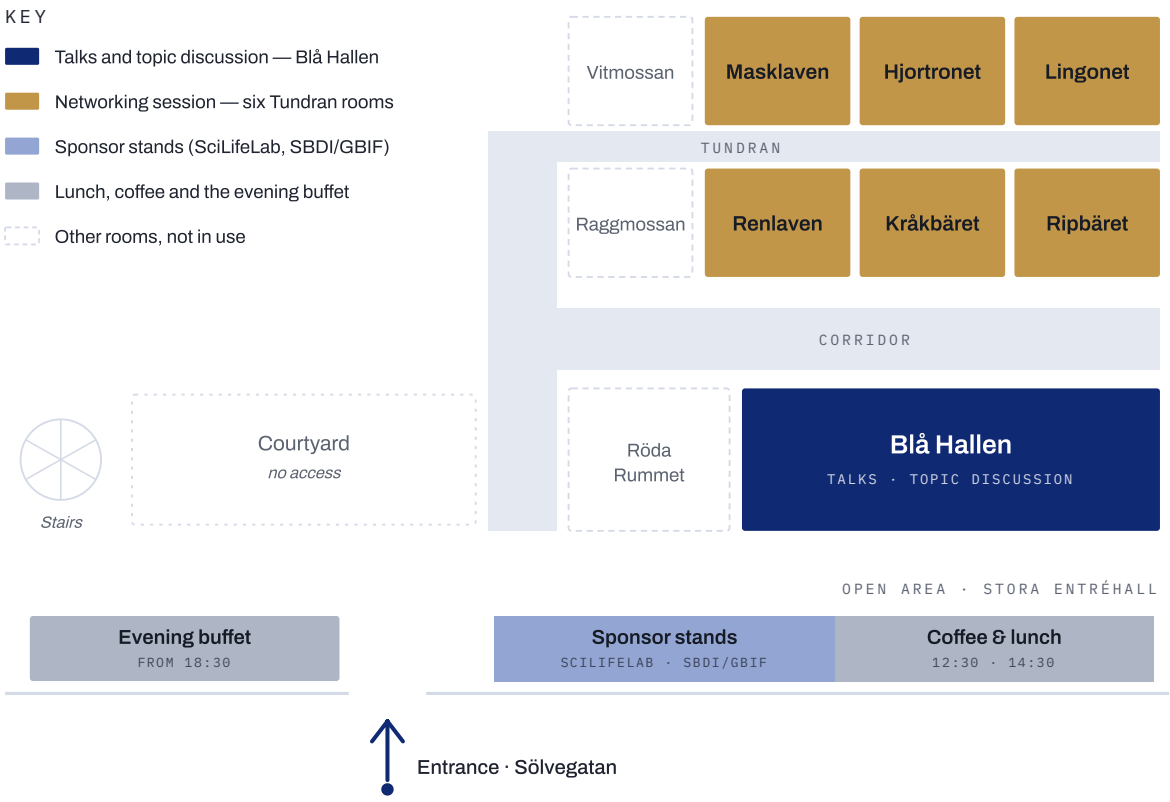
Two informal discussions, one after the other, on the themes registrants asked for most: first funding opportunities, then student supervision and teaching at Lund University together. You can stay for both. Senior colleagues from the department join as facilitators: Anna Runemark, Caroline Isaksson, Fredrik Johansson, Laura Grenville-Briggs Didymus, Magne Friberg.

MASTER MEETING – FIND YOUR PROJECT

Looking for a Master's project? This informal poster-fair session gives students the opportunity to explore ongoing research, discuss project ideas, and connect with postdocs and junior PIs to find a project that matches their interests.

Where to go

All of the day happens in the east wing of Ekologihuset, off the main entrance hall.



Schematic — rooms are shown in their real relative positions, not to scale. The sponsor, coffee, lunch and buffet blocks are tables standing in the open area between the entrance and the halls, not separate rooms.

Detailed programme

Every talk, in order, with its speaker and session chair.

A MASTER PROJECT tag marks a speaker who also offers one — follow it to the description.

08:15 – 08:45 Registration

Badges and arrival in the entrance hall

08:45 – 09:00 Introduction to the meeting

Welcome from the organising committee

09:00 – 10:30 Research session I

Chair · Gwenaëlle Deconninck

09:00	**Jan Kreider** MASTER PROJECTS	Why does complexity evolve? From social insects to multicellular organisms
09:15	**Bjørn Tore Kopperud**	How variable are speciation rates across the Tree of Life?
09:30	**Yohan Sassi**	Crossings and altitudes used by nocturnal migratory birds around the Baltic Sea
09:45	**Yedra García** MASTER PROJECTS	The community context of floral evolution
10:00	**Zsófia Tóth** MASTER PROJECT	Effect of urban environment on the sperm quality of two passerines
10:15	**Josefin Winberg**	Navigating the climate and biodiversity crises

10:30 – 11:15 Networking session

Two 20 min rounds · rotating groups of six · rooms on your badge

11:15 – 12:30 Research session II

Chair · Jan Kreider

11:15 **Rickesh Patel** MASTER PROJECT
Navigation above and below the water's surface

11:30 **Ernő Vincze** MASTER PROJECT
Food (and hormones) for thought: the effect of diet and IGF-1 hormone on the cognition of urban and rural birds

11:45 **Nicholas Planidin** MASTER PROJECT
How hybridization derived genetic variation shapes the DNA methylation of the Italian sparrow

12:00 **Danielle Clake**
Effects of habitat loss and fragmentation on population isolation and inbreeding in a grassland-specialist butterfly (*Cupido minimus*)

12:15 **Alexander Jacobsen**
Niche divergence as a GxE interaction?

12:30 – 13:30 Lunch SBDI · GBIF SWEDEN SCILIFELAB

All five SciLifeLab units at the stand

13:30 – 14:30 Topic discussion

Funding opportunities first, then student supervision and teaching

14:30 – 15:00 Coffee break SBDI · GBIF SWEDEN SCILIFELAB

15:00 – 16:15 Research session III

Chair · Danielle Clake

15:00 **Sissi Lozada Gobilard** MASTER PROJECT
Pollination and reproduction along urban gradients

15:15 **Daniela Rossoni** MASTER PROJECT
Linking developmental covariance to evolvability across levels of biological variation

15:30 **Carolina Pacheco**
Genomic legacy of near extinction and intensive management in the pink pigeon

15:45 **Anastasia Christinaki** MASTER PROJECTS
Chemical fingerprinting of spruce cell walls during fungal decomposition using O-PTIR

16:00 **Gwenaëlle Deconninck** MASTER PROJECTS
From orchids to reptiles: how an Erasmus internship opened a new world for me

16:15 – 17:15 Master meeting – Find your project

Project offers at a relaxed poster-fair mingle

17:15 – 18:15 Research session IV

Chair · Ernő Vincze

17:15 **Daisuke Aoki**
Do birds retrace their ancestors' journeys? Biogeographic history and the evolution of migration

17:30 **Karla Münzner** MASTER PROJECT
Does it pay to be toxic? When and why algae produce toxins

17:45 **Junho Yoon**
From visual symptoms to quantitative endpoints: computer vision approaches in insecticide toxicology

18:00 **Roberta Hunt** MASTER PROJECT
From wing wear to phylogeny: AI approaches to biological questions

18:15 – 18:30 Closing remarks

Raffle draw and thanks to our supporters

18:30 → Finger food buffet and drinks

Networking groups

Your two rooms are printed on your badge. Six rooms along the Tundran corridor, two rounds — you meet up to ten people and nobody twice.

ROUND 1 10:30 – 10:50

Masklaven	Anastasia · Cecilia · Junho · Yedra · Zsófia
Hjortronet	Célian · Emily · Gwenaëlle · Jan · Pablo
Lingonet	Danielle · Karla · Kevin · Masahito · Rickesh · Sascha
Renlaven	Alexander · Bjørn · Chloe · Sissi · Yohan · Zachary
Kråkbäret	Anja · Daisuke · Josefin · Nicholas · Pamela · Roberta
Ripbäret	Carolina P. · Carolina R. · Ernő · Kalle · Rachel · Rawana

ROUND 2 10:55 – 11:15

Masklaven	Anastasia · Carolina R. · Gwenaëlle · Masahito · Pamela · Yohan
Hjortronet	Alexander · Cecilia · Ernő · Nicholas · Rickesh
Lingonet	Anja · Chloe · Danielle · Emily · Junho · Kalle
Renlaven	Pablo · Rachel · Roberta · Sascha · Sissi · Zsófia
Kråkbäret	Bjørn · Jan · Josefin · Kevin · Rawana · Yedra
Ripbäret	Carolina P. · Célian · Daisuke · Karla · Zachary

Groups were set so that each room mixes divisions and keeps apart people who already share a floor, an office or a research group, while bringing together people working on close topics in *different* divisions. Six people keep the same room for both rounds — the group around them changes instead.

Topic discussion

Two informal discussions, one after the other, in Blå Hallen — 13:30 – 14:30. Senior colleagues join to answer questions; you are welcome to stay for both.

1

Funding opportunities

30 ASKED FOR THIS

ANNA RUNEMARK · LAURA GRENVILLE-BRIGGS DIDYMUS · MAGNE FRIBERG

Where to find money at this career stage, what the panels are really looking for, and how to pick the right call — from VR and Formas starting grants to Marie Skłodowska-Curie and other international grants.

2

Student supervision and teaching

24 AND 20 ASKED FOR THESE

CAROLINE ISAKSSON · FREDRIK JOHANSSON · MAGNE FRIBERG

From finding teaching opportunities at Lund University and what counts towards a teaching qualification to taking on your first Master's or PhD student and procedures you should go through.

Questions submitted through the final survey are passed to the facilitators in advance, so bring anything you would rather not ask cold.

Abstracts

In programme order, grouped by session.

A MASTER PROJECT tag marks a speaker who also offers one — follow it to the description.

RESEARCH SESSION I

09:00

Jan Kreider FUNCTIONAL ECOLOGY MASTER PROJECTS

Why does complexity evolve? From social insects to multicellular organisms

Cooperation has evolved numerous times across the tree of life, repeatedly fuelling biological complexity. For example, solitary insects have evolved to cooperate in eusocial insect colonies with a reproductive division of labour between the queen and her workers, and unicellular organisms have evolved to cooperate as multicellular organisms with trillions of cells and specialised tissues. Why such complexity evolved repeatedly in some lineages, but not others, remains poorly understood. Using a theoretical model, I show that pre-adaptations of some solitary insects or unicellular organisms can facilitate the spontaneous emergence of division of labour. Furthermore, I will present how green algae can be used to investigate whether the plastic production of multicellular groups in response to environmental triggers facilitates the evolutionary transition from unicellular to multicellular life. Rather than explaining evolution solely by the presence or absence of selection, this approach emphasises the (dis-)ability of organisms to produce specific phenotypes.

09:15

Bjørn Tore Kopperud BIODIVERSITY AND EVOLUTION

How variable are speciation rates across the Tree of Life?

Across the Tree of Life, there is strong disparity in species richness among groups. There are thousands of beetles, bats and orchids, while other groups such as lungfishes, the ginkgo tree or the tuatara reptile are relatively species poor. One approach to study such disparity is to use phylogenetic birth-death models, in which rates of speciation and extinction are allowed to change along the phylogeny. Central questions include i) how variable are speciation rates, ii) how often do speciation rates change, and iii) which types of rate change events are we able to detect? I first present a novel and more efficient statistical method for estimating diversification and rate shift histories. This enables us to explore the prevalence and nature of diversification rate shifts in more species-rich empirical phylogenies than was previously feasible (representing over 300,000 species).

09:30

Yohan Sassi EVOLUTIONARY ECOLOGY AND INFECTION BIOLOGY

Crossings and altitudes used by nocturnal migratory birds around the Baltic Sea

The Baltic Sea, in northern Europe, is part of the East Atlantic flyway, one of the 8 major bird migratory flyways. This area is regularly crossed by millions of migratory birds and has more than 600 offshore wind turbines already operating and several new offshore wind energy facilities projected. This will likely lead to an increase in the bird-human activities conflict in the coming years. A well thought spatial planning thus appears to be a crucial tool to prevent the implementation of wind turbines in areas intensively used by birds and increasing collision risks. We leveraged data collected by weather radars around the Baltic Sea, over 4 years, to map migration flows and highlight areas where conflict can arise. Specifically, we used vertical profile time series quantifying bird migratory behaviours to create movement networks between radar stations to identify where and at which altitude nocturnal migratory birds cross the Baltic Sea. We then produced maps showing migration flows across radar stations around the Baltic, highlighting the proportion of birds flying at altitudes overlapping rotors swept zones, thus providing the wind energy industry with information on sensitive areas that should be avoided.

09:45

Yedra García BIODIVERSITY AND EVOLUTION MASTER PROJECTS

The community context of floral evolution

Floral diversity arises through the interplay of pollinators, environmental conditions, and ecological interactions operating across multiple spatial and temporal scales. My research explores how ecological communities shape floral evolution by generating context-dependent selection through interactions among pollinators, antagonists, co-occurring plant species, and the abiotic environment. Using orchid communities on the Baltic island of Öland as a primary study system, I investigate how complex floral phenotypes respond to multiple selective pressures, with particular emphasis on interactions among co-flowering species that share pollinators. By combining field experiments, classical selection studies, and multivariate community analyses, my research aims to better understand how ecological interactions contribute to shaping phenotypic selection, ultimately leading to floral diversification across space and time.

10:00

Zsófia Tóth EVOLUTIONARY ECOLOGY AND INFECTION BIOLOGY MASTER PROJECT

Effect of urban environment on the sperm quality of two passerines

The decline of many bird species worldwide has been attributed to factors associated with urbanisation, such as pollution, which has been linked to reduced reproductive performance. Medical studies have shown that the percentage of sperm DNA fragmentation (a measure of DNA damage) is higher in polluted areas compared with less polluted or non-polluted areas, leading to miscarriages in women or infertility in men. Sperm DNA damage can be one of the causes of hatching failure in birds, however we do not know if the level of sperm DNA damage differ between urban and rural environments. Therefore, we studied the level of sperm DNA damage along an urbanisation gradient in Blue and Great tits in southern Sweden, expecting that the birds living in urban area would have higher sperm DNA damage. We found that Blue tits had higher sperm DNA damage than Great tits. However, contrary to our expectations, we found higher sperm DNA damage in rural environment, but only in Blue tits. Our findings are the first to show that sperm DNA damage exists in a wild bird species, and raise the question of how urban birds can maintain better sperm DNA integrity than rural birds.

10:15

Josefin Winberg MGEO

Navigating the climate and biodiversity crises

My research interests centre on land use and the interconnected challenges of climate change and biodiversity loss. With a background in physical geography, biology, and environmental science, I am interested in how human decisions shape landscapes, ecosystems, and their capacity to contribute to addressing these challenges. In my current postdoctoral research, funded by Mistra Biopath, I focus on biodiversity-related decision-making in the financial and corporate sectors. This work explores how biodiversity is understood, assessed, and incorporated into decisions by companies and financial actors, and how these decisions may ultimately influence land use and biodiversity outcomes. In this talk, I will introduce my research journey and current work, highlighting the connections between land use, biodiversity, climate, and decision-making that run through my research.

RESEARCH SESSION II

11:15

Rickesh Patel SENSORY BIOLOGY MASTER PROJECT

Navigation above and below the water's surface

Imagine hiking in the trailless wilderness of northern Sweden. Leaving your tent, most humans would be utterly lost within an hour of undirected travel. Yet, some animals, like bees and ants, can easily return to their nests, even after navigating unknown, featureless terrain. I will talk about my research investigating how arthropods overcome these navigational challenges. I will first talk about how mantis shrimp navigate their underwater environments, using spatial vectors and landmarks to return to their burrows after foraging excursions. I will then discuss my research understanding the neural basis of similar behaviors in bumblebees, using a novel lab-based behavioral assay, detailed neuroanatomy, and electrophysiology. Using what we found, I will discuss how arthropods might use spatial vectors to construct a flexible spatial representation to allow them to perform complex navigation behaviors, including making direct paths between locations of interest. Finally, I will touch on work looking at how mantis shrimp might perform vector navigation operations by comparing the detailed neuroarchitecture of the putative navigation center of their brains with that of insects, animals separated by over half a billion years of evolution.

11:30

Ernő Vincze EVOLUTIONARY ECOLOGY AND INFECTION BIOLOGY MASTER PROJECT

Food (and hormones) for thought: the effect of diet and IGF-1 hormone on the cognition of urban and rural birds

Urbanization is one of the most profound ecological changes in our time, with consequences on animal behavior and physiology. Good cognitive abilities might have greater fitness benefits in urban environment, but this may be mitigated by weaker physiological condition of more urbanized animals, as well as hormones that regulate cognitive development, such as the growth hormone IGF-1. In my project I study great tits (Parus major) in urban and rural habitats and in captivity, investigating the following questions: 1) Does the level of IGF-1 change along an urbanization gradient? 2) Does IGF-1 affect problem-solving ability, and does the strength of this relationship change with urbanization and nutritional condition? 3) Does nestling diet affect problem-solving abilities and its relationship with IGF-1 level? I am presenting preliminary results and upcoming experiments revealing the role of IGF-1 for cognition, a behavioral trait that is under selection in anthropogenic environments.

11:45

Nicholas Planidin BIODIVERSITY AND EVOLUTION MASTER PROJECT

How hybridization derived genetic variation shapes the DNA methylation of the Italian sparrow

There is growing evidence that hybridization is an important mechanism for producing novel phenotypes from standing genetic variation. However, the molecular mechanisms underlying the generation of evolutionary novelty in hybrids are poorly understood. To address this open question, we examine the pattern of admixture in the hybrid Italian sparrow (Passer italiae) and how it shapes DNA methylation. First, we determine which parental species each region of the P. italiae genome originates from. Then, we assess the degree to which DNA methylation in each ancestry block matches that of the parent species. Finally, by performing this analysis in three independently derived lineages of P. italiae, we examine the repeatability of patterns of admixture and changes in the pattern of DNA methylation during hybridization. By examining DNA methylation in the genetic mosaic of a hybrid species, we empirically test how hybridization can play a role in the evolution of novel phenotypes.

12:00

Danielle Clake BIODIVERSITY AND EVOLUTION

Effects of habitat loss and fragmentation on population isolation and inbreeding in a grassland-specialist butterfly (Cupido minimus)

Loss of biodiversity is a global concern with many contributing factors, including human-driven land cover change and the resulting habitat loss. We assess how functional connectivity and genetic erosion in pollinators is impacted by the amount and spatial arrangement of remaining habitat, focusing on the grassland specialist butterfly Cupido minimus. This species exhibits population structure at distances of less than 8 km. Based on whole genome sequencing of 200 individuals collected from 40 sites in southern Sweden we estimate functional connectivity by assessing FST and clustering. We also assess the effects of habit quantity and habitat configuration surrounding each population on level of inbreeding, approximated by runs of homozygosity and accumulation of genetic burden (genomic inference of deleterious mutations). The insights gained in this study are important both for conservation planning in areas with finite space available for habitat, and for better understanding how landscape patterns can influence functional connectivity.

12:15

Alexander Jacobsen FUNCTIONAL ECOLOGY

Niche divergence as a GxE interaction?

Competition is often evoked as the catalyst for diversification, but how exactly competition drives genetic divergence is not always understood. We posit that niche diversification can result from a gene by environment (GxE) interaction between high and low competition environments. We test this in the Trinidadian guppy Poecilia reticulata, a highly evolvable generalist fish species. Through controlled breeding and stable isotope analysis, we estimate the variance in plastic response of individuals' dietary niches when families are raised across high and low competition environments. We then probe the basis for this variation in plasticity by inferring genetic correlations with body/brain size, gut length, and boldness. Finally, we assess the evolutionary consequence of occupying different dietary niches by estimating the fitness of our fish. Our results allow us to comment on the potential of GxE interactions to drive diversification, and highlight the role that heritable traits have in an organism's competitive response.

RESEARCH SESSION III

15:00

Sissi Lozada Gobilard BIODIVERSITY AND EVOLUTION MASTER PROJECT

Pollination and reproduction along urban gradients

Urban environments can generate substantial spatial and temporal variation in pollinator assemblages, floral traits, and resource availability, yet how plant reproduction is maintained under such heterogeneous conditions remains poorly understood, particularly in tropical cities. How floral traits and pollinator assemblages vary between urban and natural populations is lacking for many native plants, particularly from dry habitats. Here, we evaluated the pollination dynamics and reproductive success of two native species Nicotiana glauca and Lupinus altimontanus across an urban gradient in the high tropical Andes. Results show a clear effect of urbanization in flower morphology, pollinator assemblages and reproductive output but in different direction and degree depending on the species. These examples contribute to a better understanding on how plants adapt to changing environments and how much anthropogenic activities affect their pollination dynamics and reproductive success.

15:15

Daniela Rossoni BIODIVERSITY AND EVOLUTION MASTER PROJECT

Linking developmental covariance to evolvability across levels of biological variation

Understanding how developmental processes generate and structure phenotypic variation is a central goal of evolutionary developmental biology. Recent comparative studies show that phenotypic covariance predicts macroevolutionary divergence as effectively as genetic covariance, suggesting that developmental processes may contribute to the organization of evolutionary variation. Here, we investigate this relationship using repeated structures in damselfly wings. Bilateral homologues provide a measure of stochastic developmental variation through fluctuating asymmetry, whereas serial homologues reveal covariance generated by shared developmental programs and their partial developmental dissociation. Using morphometrics and quantitative-genetic approaches, we estimate covariance matrices describing phenotypic variation (P), fluctuating asymmetry (FA), and serial homology (SH), and compare their patterns of integration, modularity, and evolvability within a common analytical framework. This study provides an empirical link between developmental covariance and the structure of phenotypic variation, offering new insights into how developmental architecture may influence evolutionary potential.

15:30

Carolina Pacheco EVOLUTIONARY ECOLOGY AND INFECTION BIOLOGY

Genomic legacy of near extinction and intensive management in the pink pigeon

Conservation interventions can rapidly recover population size, but whether they also restore the evolutionary potential of threatened species remains unclear. The pink pigeon (Nesoenas mayeri), endemic to Mauritius, provides a unique opportunity to address this question, having rebounded from just 9–10 known wild birds in the 1990s to ~600 today through intensive management. Using 150 whole genomes spanning 180 years, including samples collected before and after the population bottleneck, we reconstruct genomic change across the species' decline and recovery. We ask how severe bottlenecks, inbreeding, and founder representation have shaped genetic diversity during demographic rescue, and whether population growth has translated into genomic recovery. Our results show that demographic and genomic recovery can follow very different trajectories, highlighting the need for conservation strategies that consider not only census size but also effective population size, genetic diversity, and the representation of lineages.

15:45

Anastasia Christinaki FUNCTIONAL ECOLOGY MASTER PROJECTS

Chemical fingerprinting of spruce cell walls during fungal decomposition using O-PTIR

Brown-rot and white-rot fungi are major decomposers of wood in terrestrial ecosystems, yet they employ remarkably different strategies to overcome the complex structure of plant secondary cell walls. White-rot fungi rely largely on enzymatic degradation, while brown-rot fungi use iron-dependent Fenton chemistry, generating highly reactive radicals that modify wood polymers. However, where and how these chemical transformations occur within wood cell walls remains poorly understood, especially for brown-rot fungi. In this project, Optical Photothermal InfraRed (O-PTIR) spectroscopy is used to investigate chemical changes in wood during decomposition by brown- and white-rot fungi. Hyperspectral imaging enables us to map spatial and temporal changes across wood cell walls and compare decay patterns between fungal strategies. Mapping fungal-induced modifications in wood cell walls allows us to visualize changes in wood chemistry at the microscale and will guide our future studies on iron speciation, mobilization, and translocation during fungal decay.

16:00

Gwenaëlle Deconninck BIODIVERSITY AND EVOLUTION MASTER PROJECTS

From orchids to reptiles: how an Erasmus internship opened a new world for me

My research journey was not linear but surely was shaped by seizing opportunities and curiosity. Everything started with an Erasmus internship in Greece that opened the world of research ahead of me. I was inventorying orchids and developing in vitro propagation protocols. From nutrition studies, I went back to university with one aim: becoming a researcher, I wanted this to be my life. It is the 10-year anniversary of this internship. In the meantime, I have studied pollination syndromes in Ceropegia species, the impact of cold temperature on orchid development and the trophic, thermal and symbiotic components of the invasive success of Drosophila suzukii. Now in Lund, I switched to reptiles, with two main projects: questioning how behavioural thermoregulation can buffer evolution of thermal traits and what are the drivers of resilience of the wall lizard populations on Mediterranean islands. From orchids to reptiles, making it full circle.

RESEARCH SESSION IV

17:15

Daisuke Aoki EVOLUTIONARY ECOLOGY AND INFECTION BIOLOGY

Do birds retrace their ancestors' journeys? Biogeographic history and the evolution of migration

Bird migratory routes span the globe and vary remarkably among species. How did such diversity evolve? One intriguing hypothesis proposes that birds migrate by retracing the colonization history of their ancestors. This idea has been discussed for nearly a century and can explain aspects of present-day migratory routes, yet its broader evolutionary meaning remains unclear. Framing this phenomenon within evolutionary theory may help us understand how migratory routes diversify and how they may respond to ongoing environmental change. In this talk, I will explore how the migratory routes of Japanese birds may emerge from the combined effects of natural selection and historical constraints associated with ancestral breeding ranges. I will then discuss how this perspective could be extended using the well-established Willow Warbler system at Lund University, with the aim of connecting biogeography, migration ecology, and evolutionary biology.

17:30

Karla Münzner FUNCTIONAL ECOLOGY MASTER PROJECT

Does it pay to be toxic? When and why algae produce toxins

Harmful cyanobacterial blooms are a familiar sight in lakes and reservoirs around the world. They can turn clear waters bright green, produce toxins that threaten both animal and human health, and compromise drinking water supplies. Yet not every bloom becomes toxic. What determines whether a bloom produces toxins? Is there a competitive advantage in producing toxins? I explore these questions using Microcystis, a bloom-forming cyanobacterium whose populations contain both toxin-producing and non-toxin-producing strains. By growing these strains under different nutrient, light, and pH conditions, I investigate when toxic strains outgrow their non-toxic counterparts, and vice versa. Understanding the environmental conditions that favor toxic strains will help us predict not only when cyanobacterial blooms occur, but also when they become toxic. Such predictions can support timely and cost-effective measures to protect drinking water resources and protect human and ecosystem health.

17:45

Junho Yoon BIODIVERSITY AND EVOLUTION

From visual symptoms to quantitative endpoints: computer vision approaches in insecticide toxicology

Behavioral symptoms provide valuable information on how insecticides affect nervous and motor function in insects, yet they are still commonly described qualitatively using terms such as tremor, convulsion, incoordination, or paralysis. This presentation introduces computer vision as a means of converting these visual observations into quantifiable and reproducible toxicological endpoints. By tracking changes in body posture and movement, behavioral phenotypes can be objectively compared across treatments and linked to differences among insecticide classes. Two applications are presented across house flies and cotton aphids, illustrating how automated behavioral phenotyping can reveal class-specific intoxication patterns, support the interpretation of synergistic interactions in insecticide mixtures, and capture sublethal endpoints. Together, these examples illustrate the broad applicability of computer vision in insect toxicology.

18:00

Roberta Hunt BIODIVERSITY AND EVOLUTION MASTER PROJECT

From wing wear to phylogeny: AI approaches to biological questions

In this presentation I'll go over the broad strokes of a few recent projects I've worked on relating to using AI to try to answer biological questions. The first is my current project, predicting bee age through estimating wing wear using AI and shape models. The next is segmenting bees to estimate effects of pesticides on behaviour. Another was using deep metric learning to infer phylogenies from images of rove beetles. Finally (and oldest) was a project for my masters thesis on species delimitation in butterflies using latent models.

Master projects

24 projects offered by 20 participants at the poster fair, 16:15 – 17:15.

01

Célian Diblasi

BIODIVERSITY AND EVOLUTION · ONE PROJECT

Does polyploidy contribute to phenotypic difference in the context of coevolution?

To better understand the consequences of the current changes on biodiversity, one important aspect lies in the comprehension of the interaction between species. Such coevolutionary interaction exists between plants and their pollinator, which shape each other's traits. The plant species *Lithophragma bolanderi* is closely interacting with *Greya* moths, which are simultaneous pollinators and herbivores. In addition, *L. bolanderi* flowers harbor an extremely diverse set of floral morphologies and floral scent across population, and this variation is partly mediated by their pollinators. Finally, *L. bolanderi* also harbors ploidy level variation (different number of copies of the genome) across population. We hypothesize that the extreme variation in floral traits, and especially floral scent diversity, could be determined by difference in ploidy level. We thus would like to collect floral scent between different ploidy levels and compare the compounds found to investigate the role of ploidy in shaping floral scent.

02

Roberta Hunt

BIODIVERSITY AND EVOLUTION · ONE PROJECT

Phylogenetic explanations using evolutionary AI algorithms

This project will explore whether large language models can be combined with evolutionary search algorithms to identify patterns of trait evolution across phylogenetic trees. The algorithm will propose possible relationships between traits and evolutionary lineages, test those ideas against observed trait data and the phylogeny, and then use the results to generate better hypotheses in the next round. The aim is to test whether this approach can produce biologically meaningful and interpretable insights into evolutionary processes. This idea is heavily influenced by the paper "Evolving Interpretable Visual Classifiers with Large Language Models". The project could be adapted to be more on the data science side, or more on the biological side.

03

Rickesh Patel

SENSORY BIOLOGY · ONE PROJECT

Neural basis of spatial navigation in bumblebees

My research uses behavioral experimentation, targeted neurotoxin ablations of specific brain regions to understand how bumblebees construct representations of space in their brains with which to navigate. Potential Master's projects fall within the scope of these research aims and methods and can cater to the interest of the student. 10 month theses for 60 credits are prioritized, but 6 month theses may be organized as well, depending on the project.

04

Nicholas Planidin

BIODIVERSITY AND EVOLUTION | ONE PROJECT

The consequences of hybridization on the phenotypes of the Italian sparrow

Hybrids between species often have phenotypes that are more extreme than either of their parental species. This can be an important mechanism for adaptation and the formation of new species but is rarely investigated beyond first generation hybrids. To understand the consequences of hybridization over longer time scales, we are studying the Italian sparrow; a hybrid species formed from the house and Spanish sparrows around 5000 years ago. We are looking for a master's student to measure and compare phenotypes in this hybrid species relative to its parental species, with many possible phenotypes to choose from, e.g., feather coloration, song, beak shape, wing shape, nest construction, reproductive timing. For bioinformatically oriented students, there is also the possibility to compare DNA methylation among these species using long-read sequencing data.

05

Jan Kreider

FUNCTIONAL ECOLOGY | TWO PROJECTS

PROJECT 1

Does phenotypic plasticity facilitate the evolutionary transition from unicellular to multicellular life?

The project investigates whether unicellular green algae that can plastically produce multicellular groups are predisposed to evolve obligate multicellularity using experimental evolution, flow cytometry cell sorting, high-throughput cell imaging, and Bayesian phylogenetic comparative methods.

PROJECT 2

Are there molecular signatures of ancient plasticity-led evolution in multicellular green algae?

The project investigates signatures of plasticity-led evolution across eight evolutionary origins of multicellularity in green algae using RNA sequencing, comparative transcriptomics, and Bayesian phylogenetic comparative methods.

06

Yedra García

BIODIVERSITY AND EVOLUTION | TWO PROJECTS

PROJECT 1

Co-flowering community context and the maintenance of floral polymorphism in a food-deceptive orchid

Food-deceptive orchids attract pollinators without offering nectar rewards. *Dactylorhiza sambucina* is a deceptive orchid with two floral colour morphs that coexist within populations. This project will investigate whether variation in floral scent, a key trait mediating plant-pollinator interactions, parallels colour variation and whether co-flowering community composition contributes to the maintenance of floral polymorphism through pollinator-

mediated interactions. The study will combine measurements of co-flowering community composition, floral scent and reproductive success to investigate how ecological context shapes floral phenotypic variation.

PROJECT 2

The role of co-flowering communities in floral character displacement

Previous work has revealed variation in floral traits among populations of two closely related co-occurring orchid species, *Gymnadenia conopsea* and *G. densiflora*, that differ in their co-flowering communities. This project will investigate whether differences in co-flowering community composition are associated with floral trait divergence and whether such divergence could arise through pollinator-mediated selection. By combining measurements of co-flowering community composition, floral traits, and estimates of pollinator-mediated selection across sympatric and allopatric populations, the project will test whether these patterns are consistent with floral character displacement.

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07 **Daniela Rossoni** BIODIVERSITY AND EVOLUTION ONE PROJECT

Predicting long-term evolution of mandible morphology in neotropical bats

In this Master's project, we will use comparative quantitative genetic approaches and phylogenetic comparative methods to investigate how phenotypic covariance shapes the evolutionary trajectory of mandible morphology in phyllostomid bats. Using existing mandible imaging datasets and a phylogenetic framework, the project will reconstruct evolutionary divergence along branches of the phylogeny and quantify both the orientation and magnitude of morphological change.

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08 **Pamela Santana** BIODIVERSITY AND EVOLUTION ONE PROJECT

Paternity in nature: how differences in pollinator communities promote fidelity or promiscuity in plants

Plants are sessile organisms that rely on pollinators for sexual reproduction. Plants are often perceived as intrinsically "promiscuous" organisms, where receiving more pollen grains is assumed to increase reproductive success. However, pollinator behavior can fundamentally alter this expectation. Several aspects of pollination influence mating outcomes: (1) how much pollen a pollinator carries, (2) how frequently a pollinator visits a species of plant, and (3) how many pollinators species a plant shares with its neighboring plant species. Together, these factors determine whether a plant produces fruits in which most seeds share the same father (full siblings) or, alternatively, fruits in which seeds have multiple fathers (half siblings). We will be addressing this question using *Lithophragma bolanderi* as a study system. This species is particularly well suited for testing hypotheses about mating patterns because some populations are highly pollinator-specialized, whereas others are pollinator-generalists. This natural variation allows us to compare populations and test the hypothesis that more generalized pollination systems lead to higher paternal diversity within fruits. If this hypothesis

is supported, we predict that different pollinator groups will contribute unequally to effective pollen transfer, resulting in detectable differences in paternity distributions, mate diversity, and spatial patterns of gene flow among populations. To test these predictions, we sample maternal plants in the field, grow their offspring in the greenhouse, and perform genetic paternity analyses. Because pollinator assemblages are already known to differ among populations, we aim to link pollinator visitation patterns to actual mating outcomes. Seeds from focal maternal plants will be subjected to paternity analysis using microsatellite markers, allowing us to identify contributing fathers for each seed set and quantify paternal diversity, skew in male reproductive success, and pollen dispersal distances. Genetic data will be analyzed using paternity assignment software and spatial genetic tools to test relationships between pollinator composition, floral traits, and patterns of paternity within and among populations. Specifically, we will assess how pollinator identity correlates with (a) the distribution of paternal contributions, (b) pollen movement distances, and (c) male reproductive skew. By comparing these metrics across populations with contrasting ecological contexts, we can evaluate whether differences in pollination environments lead to consistent patterns of gene flow and whether such patterns may contribute to local adaptation or population divergence.

09 **Sissi Lozada Gobilard** BIODIVERSITY AND EVOLUTION DESCRIPTION TO FOLLOW

The description for this project has not been submitted yet.

10 **Zsófia Tóth** EVOLUTIONARY ECOLOGY AND INFECTION BIOLOGY ONE PROJECT

Understanding the impact of urbanization on the production of avian ejaculate

The average reproductive performance of birds has been shown to be lower in polluted areas (Eeva & Lehikoinen, 1995; Peach et al, 2008, Barton et al, 2023). Hatching success is one of the most important determinants of fitness, and the number of eggs that fail to hatch can have a serious impact on population dynamics (Ferreira et al, 2005; White et al, 2015). Hatching failure (excluding predation and abandonment) can have different proximate causes: for example, it may result from poor sperm quality, bad female condition, and/or environmental factors. Many vertebrates, such as birds, undergo annual cycles of gonad development in which the gonads must develop before breeding can commence (Davies and Deviche, 2014). It was shown that urban birds have earlier gonadal development compared with their rural conspecifics (Davies et al, 2015). It was shown in human that higher exposure to air-pollutant can cause damage to testes, characterized by histological alterations, oxidative damage and apoptosis. However, we know that urbanization has an impact on the timing and pace of gonadal development, we do not know how it can influence the process of sperm production and the quality of sperm. Therefore, the proposed project will address how urbanization and traffic related pollutants can influence the oxidative status (SOD activity, MDA level and T-AOC) and apoptosis level of the testes and the ejaculate in wild birds, and how it influences the quality of sperm (swimming velocity, progressive motility, morphology and DNA damage).

Project will be carried out on Blue and Great tits (*Cyanistes caeruleus* and *Parus major*), which makes possible that the students are working together closely during the course of their master degree, while still doing their independent research.

11 **Ernő Vincze** EVOLUTIONARY ECOLOGY AND INFECTION BIOLOGY ONE PROJECT

Levels of IGF-1 hormone and nutrition and its effects on problem-solving ability in urban and rural great tits

Environmental conditions in cities differ in many regards from nearby natural areas, providing many challenges for the animals living there. As a result, urban animals differ in many characteristics from those living in more natural habitats, including physiological traits like nutritional conditions and hormone levels, and behavioral traits like problem-solving abilities. In the below projects, we investigate potential relationships between these traits in great tits (Parus major) breeding in Malmö (urban habitat) and Skrylle nature reserve (rural habitat). We are looking for students interested in field work with birds, behavioral experiments, and/or laboratory work!

12 **Anastasia Christinaki** FUNCTIONAL ECOLOGY TWO PROJECTS

PROJECT 1

Chemical fingerprints of wood decay: finding the molecules behind iron-driven decomposition

Wood decomposition is a key process in the global carbon cycle, yet we still know surprisingly little about the small molecules fungi use to break down this highly resistant material. This project will investigate iron-active metabolites produced by brown-rot fungi during wood decomposition, combining fungal cultivation and wood-decay experiments with metabolite extraction, HPLC profiling and biochemical assays for Fe-reducing activity. The goal is to connect fungal chemical profiles with function and identify candidate molecules involved in iron-driven wood decomposition.

PROJECT 2

Chemical warfare between fungi: exploring bioactive metabolites during fungal interactions

What new chemistry emerges when two fungi compete? Fungal confrontation can activate otherwise hidden metabolic pathways, leading to the production of pigments, secondary metabolites and extracellular enzymes involved in attack and defense. This project will use dual fungal cultures, HPLC-based metabolite profiling and enzymatic assays to investigate the chemical and enzymatic responses triggered during antagonistic fungal interactions, with the aim of identifying interaction-specific metabolites and activities involved in fungal competition.

13 **Sascha Müller** MGEO ONE PROJECT

Degradation of nanoplastic by fungi

14

Kevin Young

EVOLUTIONARY ECOLOGY AND INFECTION BIOLOGY

ONE PROJECT

The impact of solar heating on high-altitude flying migratory birds

This project will involve using data collected from wind tunnel studies to provide baseline estimates, and data collected by you from a museum collection of migratory birds to quantify the impact of solar radiation on flying birds under different environmental conditions.

Full description: https://biologyedu.blogg.lu.se/the-impact-of-solar-heating-on-high-altitude-flying-migratory-birds/

15

Masahito Tsuboi

BIODIVERSITY AND EVOLUTION

ONE PROJECT

Data-driven approach to explore why evolvability is structured the way it is

In my lab, we combine field work, experiment, and analyses of big image data to study how evolvability is generated and maintained, and how evolvability today is related to evolution at long evolutionary timescales. Currently, our main model system is insect wing morphology, but we have access to data of antler morphology in deer and brain size across vertebrates. Please come talk to me if you are interested in these ideas, or more broadly in questions related to evolvability.

16

Dineshkumar Kandasamy

SENSORY BIOLOGY

ONE PROJECT

Why bark beetles partner with fungi?

The Eurasian spruce bark beetle (Ips typographus) is a devastating pest of conifers (mainly Norway spruce, Picea abies) in Europe. We study why bark beetles partner with fungi to survive and reproduce in conifers. The two main topics are 1) whether the symbioses between bark beetles and fungi are obligatory or facultative. 2) Identify the mechanisms that maintain this symbiosis.

17

Gwenaëlle Deconninck

BIODIVERSITY AND EVOLUTION

TWO PROJECTS

PROJECT 1

Competition for thermoregulation and personality traits

Sometimes, even though habitat characteristics suggest that species should be able to reach their thermal preference, they don't (i.e., their body temperature is always lower than their thermal preference). One way to explain it would be competition with other species that exploit a similar thermal niche The project questions if personality traits of species can explain their realised thermal niche with the hypothesis that species that are more mobile, exploratory, risk-taking, and aggressive would more likely be better able to reach their thermal preference compared to species on the other side of this personality spectrum. We use reptiles as a model

as data on their thermal niche are available and the project will involve collection of data on reptile personality traits such as movement activity, exploration, risk-taking, sociability and aggression (the big five of behavioural ecology) and use of comparative methods.

PROJECT 2

Nest site selection in the sand lizard Lacerta agilis

How female sand lizards select their nesting site, in particular related to temperature, and how it impacts hatching success. Nesting site is among the most limiting factor for sand lizards, because they need sandy slopes that are sun exposed to develop successfully. The project includes microclimatic measurement (specifically temperature) of potential nesting sites in the field with comparison of suitable and unsuitable sites and also aims to relate these measurements to the hatching success. The project could also include laboratory experiments with thermal gradients for female to select oviposition sites (this would be dependent on obtention of breeding permits which might be delicate for sand lizards as they are threatened in Sweden). This is more of a conservation project.

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18 **Karla Münzner** FUNCTIONAL ECOLOGY ONE PROJECT

When cyanobacteria blooms turn toxic: growth dynamics of toxic and non-toxic Microcystis spp. strains

The toxic cyanobacteria Microcystis sp. is one of the most common toxin-producing cyanobacterium in freshwater lakes. It produces the toxin microcystin, which to date has an unresolved function, but is highly toxic to humans and other mammals when consumed. Populations of Microcystis consist of two types of unique cell types, or genetic strains: toxic strains that produce the toxin microcystin, and non-toxic strains that cannot produce any toxins. Producing toxins carries a cost in form of energy, but it remains unclear how high this cost is for toxic Microcystis strains compared non-toxic ones, and under which environmental conditions this cost is most pronounced. Lake water pH levels and nitrogen concentrations have been linked to a dominance of toxic Microcystis strains during bloom formation, but lab studies investigating the growth dynamics of toxic and non-toxic strains have yielded contradictory results, likely due to the low number of strains used. This master thesis' aim is to investigate the intraspecific variations of growth between toxic and non-toxic Microcystis strains under different environmenal factors and identify environmental conditions that favour toxic strains over non-toxic ones. For this, we use multifactorial laboratory experiments using a wellplate set-up that allows for high-throughput analysis of many strains at once. If you are interested in research about when and why cyanobacterial blooms become toxic and study those questions in a laboratory setting using algae cultures, please contact me for further information!

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19 **Rachel Steward** BIODIVERSITY AND EVOLUTION ONE PROJECT

How plant chemistry and microbial associations govern host use by thistle specialist flies

This project compares chemical and nutritional data among infested and uninfested thistle buds to try to understand the drivers underlying specialization and parallel diet shifts in divergent host races of the peacock fly Tephritis conura. We will further test whether the chemical landscape of the plants covaries with the microbial associations of the flies to learn whether endosymbionts play an important role in mediating this interaction.

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20 **Katharina Ruthsatz** EVOLUTIONARY ECOLOGY AND INFECTION BIOLOGY TWO PROJECTS

PROJECT 1

How do early-life heat stress and prenatal stress hormones shape gut microbiome development in wild birds?

The gut microbiome may influence how young animals develop and respond to environmental challenges, but little is known about how warming and prenatal conditions jointly shape its development in wild birds. This project will investigate how experimental nest heating and prenatal exposure to corticosterone affect gut microbiome diversity and composition in great tit (Parus major) nestlings. By sampling the same nestlings at different ages, we will examine how the microbiome changes during development and whether these treatments alter its developmental trajectory. The project combines fieldwork with wild birds, sample collection, molecular laboratory work and analysis of microbiome sequencing data. The balance between laboratory work and bioinformatics can be adapted to the student's interests.

PROJECT 2

Are differences in the gut microbiome linked to how young birds cope with heat?

Why do some nestlings cope better with heat than others, and could their gut microbes help explain this variation? This project will investigate growth and physiological responses to experimental nest heating and prenatal corticosterone exposure in great tit (Parus major) nestlings. Combining physiological measurements with microbiome data from a complementary project, we will examine whether gut microbiome diversity and composition are associated with differences in metabolic and thermoregulatory performance. The project combines fieldwork with wild birds, physiological measurements and statistical analysis of host–microbiome relationships. The specific physiological measurements and scope will be adapted to feasibility and the student's interests.

Sponsors

Two of our sponsors are with us on the day — come and talk to them.

SciLifeLab

SOPHIE HELLSTRAND · SITE COORDINATOR · AT THE STAND ALL DAY

SciLifeLab is Sweden's national research infrastructure for life science: nine capability areas, from genomics and proteomics to imaging, structural biology and bioinformatics, open to researchers from academia, industry and healthcare. SciLifeLab Lund launched in 2022 as a cross-faculty initiative between Medicine, Science and Engineering, and now runs eleven units.

Five of those units join us **over lunch**, 12:30 – 13:30:

NBIS Katja Kozjek	National Bioinformatics Infrastructure Sweden — computational and bioinformatics support, from experimental design to analysis.
LBIC — Lund University Bioimaging Centre Two imaging specialists	Advanced imaging equipment and techniques, from micro to macro — two specialists here, covering microscopy and in vivo imaging.
Training Hub Jill Jaworski	Courses and workshops in bioinformatics, data analysis, open science and machine learning, for more than 2 500 participants a year. Bring a question and browse the training portal with her.
CTG — Clinical Genomics Lund Julia Bräunig	Genomic technologies for translational projects and for bringing new diagnostic assays into healthcare.
Structural Proteomics Anton Sellerberg	Mass spectrometry for protein interactions and conformational dynamics, within the Integrated Structural Biology platform.

USEFUL IF YOU — have an experiment that needs a technology or an analysis your own lab cannot run — staff scientists support you from study design to data handling.

ONLINE — scilifelab.se · training.scilifelab.se

Swedish Biodiversity Data Infrastructure

Home of GBIF Sweden

SBDI and GBIF Sweden

RASA BUKONTAITE · PROJECT COORDINATOR, SBDI · AT THE STAND UNTIL 15:00

SBDI — the Swedish Biodiversity Data Infrastructure — is Sweden's national research infrastructure for biodiversity data, funded by the Swedish Research Council. GBIF Sweden is part of it: the national node of the Global Biodiversity Information Facility, hosted by the Swedish Museum of Natural History since 2003. Between them they connect Swedish collections and observations to the largest open database of species occurrence records in the world — all of it free to search, download and reuse.

USEFUL IF YOU	need occurrence data for a species, region or time period — or want your own dataset published and citable, with a DOI.
ASK RASA ABOUT	finding and cleaning data, Darwin Core, licensing, FAIR and CARE principles, and the GBIF API for programmatic access.
WORTH KNOWING	**Data Fika** — a free monthly online session, Fridays 11:00–11:45, running to June 2027. Ten sessions take you from a first download to publishing your own data: a short talk plus hands-on practice. Email gbif@nrm.se with "Data Fika" in the subject to be added.
ONLINE	gbif.se · gbif.org · biodiversitydata.se

WE ALSO THANK

KULTUREN

Kulturen

50 MUSEUM TICKETS · ONE FOR EVERY REGISTERED PARTICIPANT

A cultural history museum in the middle of Lund, built around an open-air site of historic houses from across southern Sweden — homes of different periods and different social classes, set among gardens. Indoors there are exhibitions on cultural history, archaeology and photography, with guided tours, talks and seasonal celebrations through the year, plus a restaurant, café and shop. Supported by Lund Municipality and Region Skåne.

ONLINE	kulturen.com

Tropikariet

5 ENTRY TICKETS · RAFFLE PRIZE

A tropical zoo and aquarium in Helsingborg, a short trip up the coast. The route takes you through a moonlit rainforest, down to a coral reef off the Maldives and past a group of lemurs from Madagascar, with conservation and education work running alongside the exhibits. Open Tuesday to Sunday, 11:00 – 17:00.

ONLINE tropikariet.com

Food and breaks

Three kitchens keep the day going — all of them local to Lund.

Le Mani

COFFEE · 10:30 AND 14:30

The café just along the entrance hall from us, here in Ekologihuset. They serve both breaks — coffee and something to eat during the first networking session in the morning, and again at the afternoon break before Research session III.

Notch Matstudio

LUNCH · 12:30 – 13:30

A vegetarian and vegan food studio on Stora Gråbrödersgatan in central Lund, cooking seasonal food they describe as *smakrikt i säsong* — flavourful, in season. They are making the lunch wraps.

ONLINE notchmatstudio.se

Mersmak

EVENING BUFFET · FROM 18:30

The kitchen behind Mötesplats CRC, catering the vegetarian finger-food buffet and drinks that close the day.

Lunch and the evening buffet are both vegetarian. If you have an allergy or a dietary need we have not already noted, tell us at registration and we will pass it on.

Contact: Gwenaëlle Deconninck